

2023 Year 6 Term 3 Common Test H2 Physics Solutions

- 1 D At midway between the charges, the directions of E_Q and E_R are both towards the right. Taking right as positive,

$$\begin{aligned}
 E &= \frac{q_P}{4\pi\epsilon_0 r^2} + \frac{|q_Q|}{4\pi\epsilon_0 r^2} \\
 &= \frac{q_P + |q_Q|}{\pi\epsilon_0 d^2} \quad \left(\because r = \frac{d}{2} \right) \\
 &= \frac{8.6 \times 10^{-9} + 4.5 \times 10^{-9}}{\pi (8.85 \times 10^{-12}) (9.5 \times 10^{-2})^2} \\
 &= 5.2 \times 10^4 \text{ N C}^{-1}
 \end{aligned}$$

- 2 A By Newton's second law of motion, taking down as positive,

$$F_{\text{net}} = ma$$

$$qE = ma_y$$

$$\frac{qV}{d} = ma_y$$

$$a_y = \frac{qV}{md} \quad \dots \quad (1)$$

Considering horizontal motion,

$$s_x = u_x t$$

$$t = \frac{s_x}{u_x}$$

$$= \frac{l}{v} \quad \dots \quad (2)$$

Considering vertical motion,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$= \frac{1}{2} a_y t^2 \quad (\because u_y = 0) \quad \dots \quad (3)$$

Substituting (1) and (2) into (3),

$$s_y = \frac{1}{2} a_y t^2$$

$$= \frac{1}{2} \left(\frac{qV}{md} \right) \left(\frac{l}{v} \right)^2$$

$$= \frac{qVl^2}{2mdv^2}$$

$$\begin{aligned}
 V_{PQ} &= \left(\frac{s_y}{d} \right) V \\
 &= \left(\frac{qVl^2}{2mdv^2} \right) V \\
 &= \frac{qV^2 l^2}{2md^2 v^2}
 \end{aligned}$$

- 3 B X is LDR, Y is thermistor.
p.d. across LDR increases as its resistance has increased (potential divider principle).
Overall, there is no change in current as it is possible that the amount of increase in resistance of LDR equals that of decrease in resistance of thermistor.

- 4 C As R increases, current decreases.
 $P_r = I^2 r$
Hence P_r decreases.

- 5 B $\frac{1}{R_{XY}} = \frac{1}{R_{PQR}} + \frac{1}{R_{PST}} + \frac{1}{R_{PT}}$
 $\frac{1}{R_{XY}} = \frac{1}{1+2} + \frac{1}{2+4} + \frac{1}{3}$
 $R_{XY} = 1.2 \, \Omega$
 Option A: $R_{XY} = 1.9 \, \Omega$
 Option C: $R_{XY} = 1.6 \, \Omega$
 Option D: $R_{XY} = 2.0 \, \Omega$

- 6 D By Left Hand rule, the direction of the magnetic field is out of the page.

$$\begin{aligned}
 Bev &= \frac{mv^2}{r} \\
 B &= \frac{mv}{er} \\
 &= \frac{9.11 \times 10^{-31} \times 1.46 \times 10^7}{1.6 \times 10^{-19} \times 0.035} \\
 &= 2.38 \times 10^{-3} \, \text{T}
 \end{aligned}$$

- 7 B Gain in KE = loss in GPE

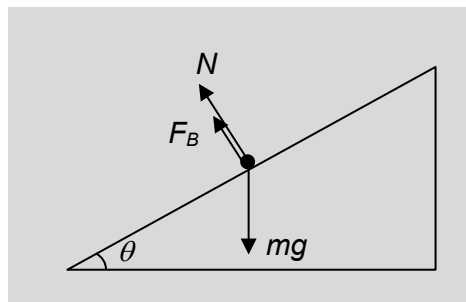
$$\begin{aligned}
 \frac{1}{2}mv^2 &= mg \left(\frac{L}{2} \times \sin \theta \right) \\
 v &= \sqrt{gL \sin \theta}
 \end{aligned}$$

$$\text{magnetic force, } F_B = Bqv = Bq\sqrt{gL \sin \theta}$$

Resolving forces perpendicularly to the plane,

$$N + Bq\sqrt{gL \sin \theta} = mg \cos \theta$$

$$N = mg \cos \theta - Bq\sqrt{gL \sin \theta}$$



- 8 **B** Using Right Hand Rule, the direction of induced e.m.f. is from P to O.
Using OP line as the reference, when the disc rotates in anticlockwise direction, the electrons will be moving to the left and hence conventional current is to the right. Using Fleming's Left Hand Rule, there will be an upward magnetic force acting on the electrons, causing them to accumulate near P. Therefore, O will have a higher potential.
- 9 **C** When the south pole of the magnet is approaching the loop, according to Lenz's law, the direction of the magnetic field induced is such that it will produce a south pole at the top of the loop to repel the magnet. Using right hand grip rule, the direction of the induced current is clockwise.

When the magnet is leaving the loop, the direction of the magnetic field induced is such that it will produce a south pole at the bottom of the loop to attract the magnet. Using right hand grip rule, the direction of the induced current is anticlockwise.

- 10 **D** $P = I^2 R$
 $I_{dc}^2 \times R = I_{rms}^2 \times 4R$
 $I_{rms} = \sqrt{\frac{I_{dc}^2}{4}} = \sqrt{\frac{(2.4)^2}{4}} = 1.2 \text{ A}$
 $I_0 = \sqrt{2} I_{rms} = 1.7 \text{ A}$
- 11 **D** $I_P V_P = I_S V_S = I_S (I_S R)$
 $I_P = \frac{I_S (I_S R)}{V_P} = \frac{1.3(1.3 \times 15)}{12} = 2.1 \text{ A}$

- 12 **A** $\max KE = hf - \Phi$
 Maximum kinetic energy and hence maximum speed of photoelectrons is independent of the intensity of the light, but dependent on frequency f and work function of the metal Φ

- 13 **D** $E = \frac{hc}{\lambda}$
 $\lambda = \frac{hc}{E}$
 $= \frac{hc}{\text{eV}}$
 $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(1.60 \times 10^{-19})(30 \times 10^3)}$
 $= 4.14 \times 10^{-11} \text{ m}$

- 14 **A**

- 15 **C** $A_0 = 673 - 17 = 656$
 After 2 half-lives, activity will be one-quarter of A_0 .
 $A = \frac{A_0}{4} = \frac{656}{4} = 164 \text{ counts per min}$
 Reading = $164 + 17 = 181 \text{ counts per min}$

16 (a) (i) $W = \Delta U$
 $W_1 = 0 \text{ J}$ A1

(ii) $W = \Delta U$
 $W_2 = \frac{q_1 q_2}{4\pi\epsilon_0 r} \quad (\because U_i = 0)$

$$= \frac{(4.5 \times 10^{-12})(4.5 \times 10^{-12})}{4\pi(8.85 \times 10^{-12})(1.5)}$$

$$= 1.21 \times 10^{-13} \text{ J}$$
 A1

Alternatively,
 $W = \Delta U = q\Delta V$

$W_2 = q_2 V_B \quad (\because V_i = 0)$

$$= q_2 \left(\frac{q_1}{4\pi\epsilon_0 r} \right)$$

$$= \frac{(4.5 \times 10^{-12})(4.5 \times 10^{-12})}{4\pi(8.85 \times 10^{-12})(1.5)}$$

$$= 1.21 \times 10^{-13} \text{ J}$$
 A1

Marker's comments: Some students do not know that p is the symbol for the prefix pico and is equivalent to 10^{-12} . A handful of students used the wrong formula for electric potential energy or got the separation wrong. Several students used the Coulomb constant $k_e = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ which is discouraged as it has a slightly different value from that computed using the value of permittivity of free space ϵ_0 given in the list of data.

(iii) $W = \Delta U$
 $W_3 = \frac{q_1 q_3}{4\pi\epsilon_0 r} + \frac{q_2 q_3}{4\pi\epsilon_0 r}$ M1

$$= \frac{q_3}{4\pi\epsilon_0 r} (q_1 + q_2)$$

$$= \frac{(-2.4 \times 10^{-12})(4.5 \times 10^{-12} + 4.5 \times 10^{-12})}{4\pi(8.85 \times 10^{-12})(1.5)}$$

$$= -1.29 \times 10^{-13} \text{ J}$$
 A1

Alternatively,
 $W = \Delta U = q\Delta V$

$W_3 = q_3 V_C \quad (\because V_i = 0)$

$$= q_3 \left(\frac{q_1}{4\pi\epsilon_0 r} + \frac{q_2}{4\pi\epsilon_0 r} \right)$$

$$= \frac{q_3}{4\pi\epsilon_0 r} (q_1 + q_2)$$
 M1

$$= \frac{(-2.4 \times 10^{-12})(4.5 \times 10^{-12} + 4.5 \times 10^{-12})}{4\pi(8.85 \times 10^{-12})(1.5)}$$

$$= -1.29 \times 10^{-13} \text{ J}$$
 A1

Marker's comments: Some students forgot to include the negative sign for q_3 . Several students made mistakes when rounding the final answer to two or three significant figures.

(b) $\Delta U = W_1 + W_2 + W_3$

$$U_f - U_i = 0 + 1.21 \times 10^{-13} + (-1.29 \times 10^{-13})$$

$$U = -8 \times 10^{-15} \text{ J} \quad (\because U_i = 0)$$

B1

Alternatively,

$$\Delta U = \frac{q_1 q_2}{4\pi\epsilon_0 r} + \frac{q_1 q_3}{4\pi\epsilon_0 r} + \frac{q_2 q_3}{4\pi\epsilon_0 r}$$

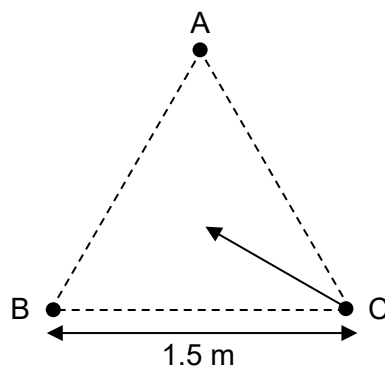
$$U_f - U_i = \frac{q}{4\pi\epsilon_0 r} (q + 2q_3) \quad (\because q_1 = q_2 = q)$$

$$U = \frac{4.5 \times 10^{-12}}{4\pi(8.85 \times 10^{-12})(1.5)} [4.5 \times 10^{-12} + 2(-2.4 \times 10^{-12})] \quad (\because U_i = 0)$$

$$= -8.09 \times 10^{-15} \text{ J}$$

B1

(c) (i)



Arrow should be pointed towards mid-point of AB.

B1

Marker's comments: The point of application of force must coincide with point C. Some students drew more than one arrow without clear label.

(ii) total initial energy = total final energy

$$E_i = E_f$$

$$\cancel{K_i} + U_i = K_f + U_f$$

$$K_f = U_i - U_f$$

$$\frac{1}{2}mv^2 = -q_3\Delta V \quad (\because \Delta U = q\Delta V)$$

$$= -q_3(V_M - V_C) \quad (\text{where M is the midpoint of AB})$$

$$= -q_3 \left[\left(\frac{q_1}{4\pi\epsilon_0 r_{AM}} + \frac{q_2}{4\pi\epsilon_0 r_{BM}} \right) - \left(\frac{q_1}{4\pi\epsilon_0 r_{AC}} + \frac{q_2}{4\pi\epsilon_0 r_{BC}} \right) \right] \quad \text{M1}$$

$$= -q_3 \left(\frac{q}{2\pi\epsilon_0 r} \right) \quad \left(\because q_1 = q_2 = q, r_{AM} = r_{BM} = \frac{r_{AC}}{2} = \frac{r_{BC}}{2} \text{ and } r_{AC} \right)$$

$$\frac{1}{2}mv^2 = -\frac{q_3 q}{2\pi\epsilon_0 r}$$

$$v = \sqrt{-\frac{q_3 q}{\pi\epsilon_0 m r}}$$

$$= \sqrt{-\frac{(-2.4 \times 10^{-12})(4.5 \times 10^{-12})}{\pi\epsilon_0 (1.7 \times 10^{-20})(1.5)}}$$

M1

$$= 3.90 \times 10^3 \text{ m s}^{-1}$$

A1

Alternatively,

gain in kinetic energy = loss in electric potential energy

$$K_f = U_i - U_f$$

$$\frac{1}{2}mv^2 = q_3V_C - q_3V_M \quad (\text{where M is the midpoint of AB})$$

$$= W_3 - q_3 \left(\frac{q_1}{4\pi\epsilon_0 r_{AM}} + \frac{q_2}{4\pi\epsilon_0 r_{BM}} \right)$$

$$= W_3 - \frac{q_3 q}{\pi\epsilon_0 r} \quad \left(\because q_1 = q_2 = q, r_{AM} = r_{BM} = \frac{r}{2} \text{ and } r = 1.5 \text{ m} \right) \quad \text{M1}$$

$$\frac{1}{2}(1.7 \times 10^{-20})v^2 = -1.2948 \times 10^{-13} - \frac{(-2.4 \times 10^{-12})(4.5 \times 10^{-12})}{\pi(8.85 \times 10^{-12})(1.5)} \quad \text{M1}$$

$$v = 3.90 \times 10^3 \text{ m s}^{-1}$$

A1

Marker's comments: Some students attempted to solve the question by considering forces and applying kinematic equations but to no avail. Kinematic equations can only be applied when acceleration is constant. A handful of students equate change in kinetic energy to change in electric potential energy which is wrong. A small number of students used the modulus function which is unnecessary with better grasp of concept. Several students used the value in (b) which is incorrect as both q_1 and q_2 remain fixed after the release of q_3 . Quite a few students appeared to equate the loss in electric potential energy directly to the value in (a)(iii) without elaboration. No credit was given.

- (iii) q_3 starts at rest and accelerates towards midpoint of A and B moving along a straight line (perpendicular bisector of A and B) B1

speed of q_3 increases (from zero) to a maximum when it is midway between A and B

q_3 then decelerates until it comes to a stop at a distance of 1.5 m from q_1 and q_2 (on the opposite side of the line joining q_1 and q_2) B1

q_3 then decelerates until it comes to a stop at C' where C' is the reflection of C about line AB

q_3 then accelerates towards midpoint of A and B and then decelerates to a stop at C. The motion then repeats.

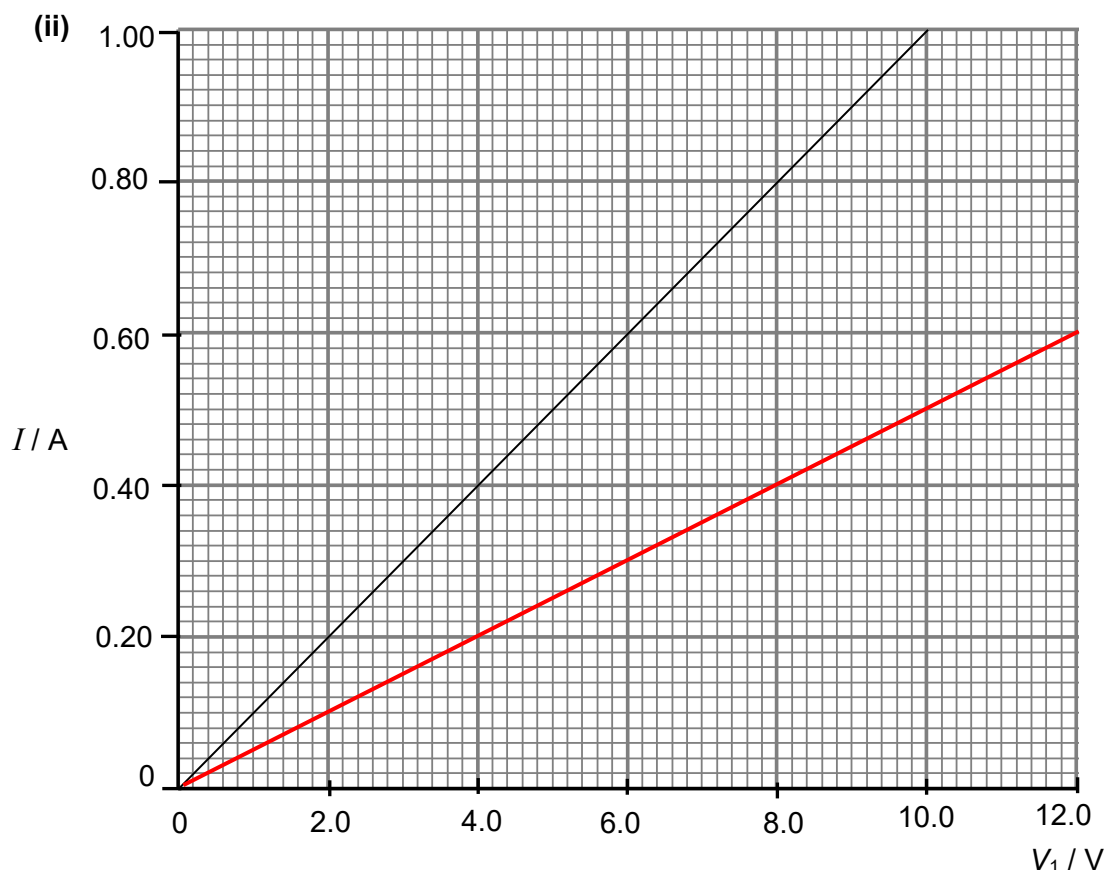
Marker's comments:

- Many students gave lengthy explanations instead of describing the motion of q_3 . This was not what was asked for.
- Many students did not use the precise term 'accelerate' but instead used vague terms such as 'move' or 'travel'. Other responses did not specify the point q_3 accelerates towards but merely stated that it will accelerate towards line AB which consists of infinitely many points.
- Some students incorrectly wrote that q_3 will speed up before slowing down to a stop at the midpoint of AB because the net force there is zero.
- Many students did not state the position where the charge comes to rest, and some wrote that the charge would end up at infinity.
- The marker accepted the use of the term 'mirror image'.
- Students who mentioned that the acceleration increases (or decreases) when moving towards (or away from) the midpoint of AB will not score the full credit as acceleration first increases then decreases to zero when moving from C to the midpoint of AB. For this same reason, the charge does not undergo simple harmonic motion which was included in many other answers.

- 17 (a) (i) The electric current passing through the resistor is proportional to the potential difference across it. A1

Marker's comments: Students need to note that saying $R = V/I$ is basically defining resistance (and not what the question is asking for). Even for non-ohmic conductor, R is still equal to V/I , and for non-ohmic cases, R is not a constant.

It is insufficient to simply say that a resistor is ohmic when it obeys Ohm's Law. Ohm's law is to be stated.



Marker's comments: Since resistivity of Y is double that of X and the dimensions are the same, it implies that resistance of Y is double that of X. Choosing any two points on the graphs, resistance can be calculated.

Students need to note that graph is a I against V and not V against I .

$$(iii) \quad R_X = \frac{8.0}{0.8} = 10 \, \Omega$$

$$R_Y = 2 R_X = 20 \, \Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{12}{10} + \frac{12}{20}$$

$$= 1.8 \, \text{A}$$

M1

A1

OR

When V is 12 V, $I_Y = 0.6 \, \text{A}$. $I_X = 1.2 \, \text{A}$ since R_X is half of R_Y .

B1

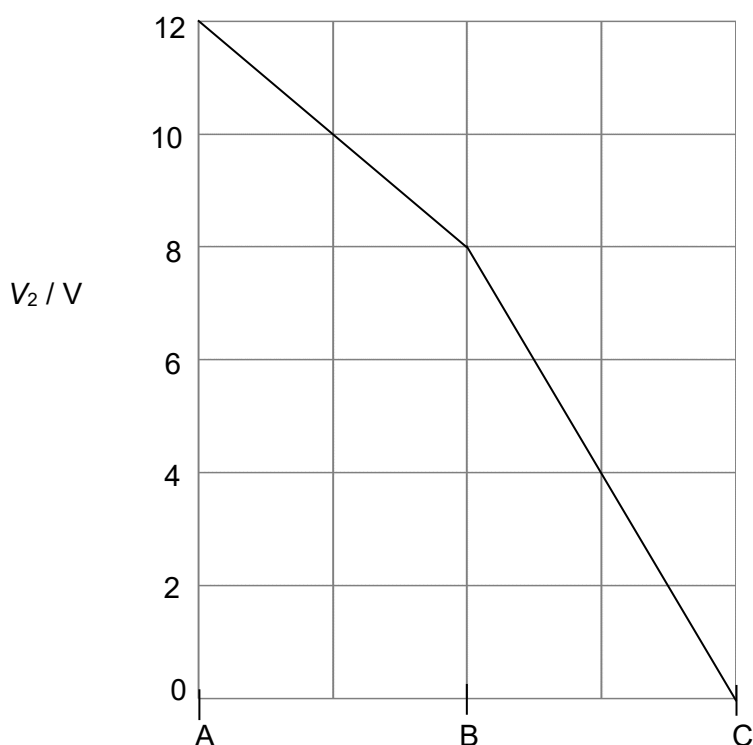
$$I = I_X + I_Y = 1.2 + 0.6 = 1.8 \, \text{A}$$

B1

Marker's comments: Some students assumed that resistance of X and Y are $1 \, \Omega$ and $2 \, \Omega$ respectively, which is not correct.

Some students did not read the question carefully and solve for the current in X instead of calculating the total current.

(iv)



1 mark: A to B is $\frac{1}{2}$ steepness compared to B to C

1 mark: correct values of V

$$\text{p.d. across X} = \frac{R_X}{R_X + R_Y} \times E = \frac{R_X}{R_X + 2R_X} \times 12 = 4.0 \text{ V}$$

Marker's comments: The question mentioned X and Y are "uniform wires" – hence the variation of p.d. with length should be linear (straight line).

(b) (i) e.m.f. of cell E is greater than p.d. across X (V_{AB})

A1

Marker's comments: Stating that "resistance of X is too small" is not sufficient. The p.d. across X depends on the resistance of X **as well as Y**.

$$(ii) \quad E = \frac{R_{JC}}{R_X + R_Y} \times V_{AC} = \frac{0.625 \times 20}{10 + 20} \times 6.0 = 2.5 \text{ V}$$

A1

Marker's comments: Other methods of solving are allowed as long as it is conceptually correct.

(c) (i) Cell E's internal resistance does not affect answer in (ii) as no current flows across its internal resistance/cell E (no potential difference). B1

Internal resistance of 6.0 V battery causes answer in (ii) to be less because A1

$$E = \frac{0.625 R_Y}{R_X + R_Y + r} \times 6.0 \text{ where } r \text{ is internal resistance of 6.0 V battery.} \quad \text{M1}$$

OR

p.d. across BC will be smaller/less than 4 V,
answer in (ii) will be smaller.

M1

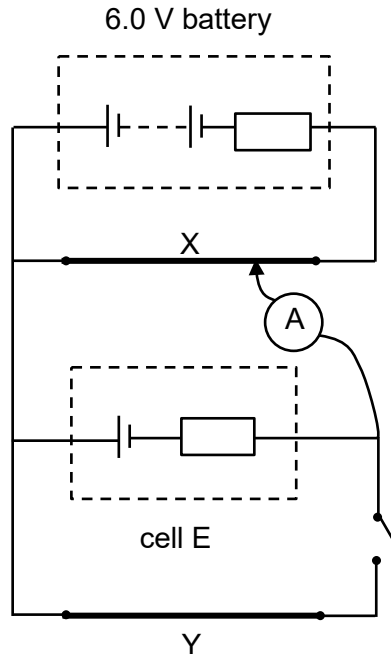
A1

Marker's comments: There were several vague answers given, e.g., "when the battery has internal resistance, the internal resistance *"takes up"* some of the p.d." hence less p.d. for the balance length. Students should avoid using such terms.

A number of students misinterpreted the question as what changes will take place to the *balance length*. In this question, the **balance length is already determined**, but the e.m.f. calculated in (b)(ii) is based on the assumption there is no internal resistance. Hence, the question is asking what the true value of the e.m.f. is when internal resistance cannot be ignored.

(ii)

B1

**For information:**

When switch is open:

$$E = \left(\frac{L_1}{R_x + r_1} \right) (6) \quad \text{where } L_1: \text{balance length, } r_1: \text{internal resistance of 6.0 V battery.}$$

When switch is closed:

$$\frac{R_y}{R_y + r} E = \left(\frac{L_2}{R_x + r_1} \right) (6) \quad \text{where } L_2: \text{new balance length, } r: \text{internal resistance of cell E.}$$

Dividing the two equations, we have $\frac{R_y}{R_y + r} = \frac{L_2}{L_1}$.

Since R_y is already known, r can then be calculated.

- 18 (a) The magnetic flux density of a magnetic field is numerically equal to the force per unit length per unit current acting on a long straight current-carrying conductor placed at right angles to the magnetic field.

B1

B1

Marker's comments: "force per unit current" brings out the meaning that B is proportional to F/I while "a conductor carrying a unit current" may mean either F/I or $F \times I$. Hence, only "per unit current" is accepted.

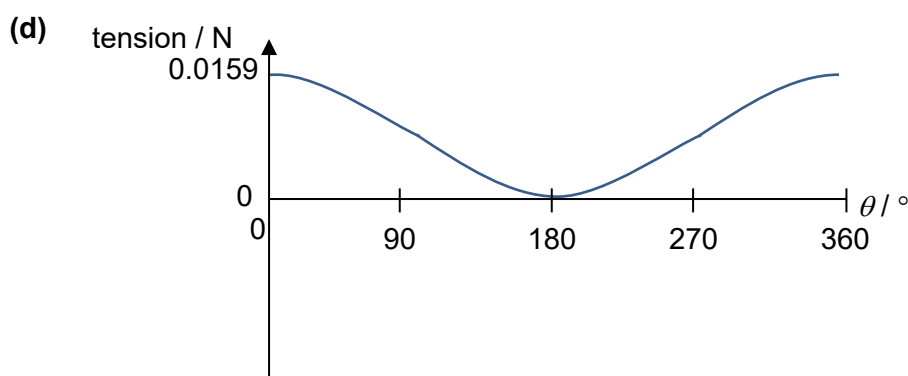
Students should be careful in using the pronoun "it" as it is sometimes unclear which quantity "it" is referring to. Instead of "placed perpendicular to **it**", students should write "placed perpendicular to the magnetic field".

(b) $B = \frac{\mu_0 NI}{2r}$
 $= \frac{4\pi \times 10^{-7} \times 400 \times 3.65}{2 \times 0.125}$ B1
 $= 7.34 \times 10^{-3} \text{ T}$ A1

(c) (i) tension $= 0.808 \times 10^{-3} \times 9.81 = 7.93 \times 10^{-3} \text{ N}$ A1

(ii) magnetic force $= BIL \sin \theta$ M1
 $= 2 \times 7.34 \times 10^{-3} \times 6.00 \times 0.090 \times \sin 90^\circ$
 $= 7.93 \times 10^{-3} \text{ N}$ A1

Marker's comments: Only the lower horizontal side of the wire frame will produce a vertical force. The two vertical sides will produce horizontal forces (to the right and to the left respectively).



Correct x-intercept B1 (Value of T at 0° is not required)
 Correct shape (cosine graph) and period B1

Marker's comments: ECF is given for students who have calculated (c)(ii) wrongly. Students must be mindful of the details of the shape of the graph, e.g., the gradient at the turning points should be zero.

(e) $L = 0.200 + (0.050 - 0.042) = 0.208 \text{ m}$ C1
 maximum magnetic torque $= BILd$ M1
 $= 2 \times 7.34 \times 10^{-3} \times 6.00 \times 0.208 \times 0.090$
 $= 1.65 \times 10^{-3} \text{ N m}$ A1

Marker's comments: Students who used $L = 0.200 \text{ m}$ are deducted one mark. Students who mixed up L and x in the formula for moment ($= BIL \cdot x$) are not given any credit even though they obtained the correct numerical value for the answer.

- 19 (a) (i) Using Fleming's Left Hand rule, the electrons will be pushed towards P. M1
 Hence, Q will be at the higher potential. A1

Marker's comments:

- In using Fleming's Left Hand rule, conventional current is taken to be upwards as the electrons in the rod are moving downwards when the rod is moving downwards.
- As there are free electrons in conductors, the force pushes the electrons towards P rather than the positive charges towards Q. Hence, positive charges should not be mentioned.
- While there is induced e.m.f. in the rod, there is no induced current as there is no complete circuit for current to flow. Hence, any answer that mention Lenz's law or induced current is not accepted.
- Do note that the use of Fleming's Right Hand rule to explain the direction of induced current is not accepted in A-levels.

(ii) $L = 0.30 \cos 20^\circ$ C1

$$E = BLv$$

$$0.013 = (0.024)(0.30 \cos 20^\circ)v$$
 M1

$$v = 1.92 \text{ m s}^{-1}$$
 A1

Marker's comments: Common mistakes:

- Forget to resolve v to be perpendicular to rod.
- Using $\cos 70^\circ$ or $\cos 30^\circ$ instead of $\cos 20^\circ$.
- Using $E = BLv$ but treating E as electric field strength instead of e.m.f.

(b) (i) By Faraday's law of electromagnetic induction, since there is no change in magnetic flux density and area of the wire loop within the magnetic field / there is no change in the magnetic flux linkage, there is no e.m.f. induced. B1

OR

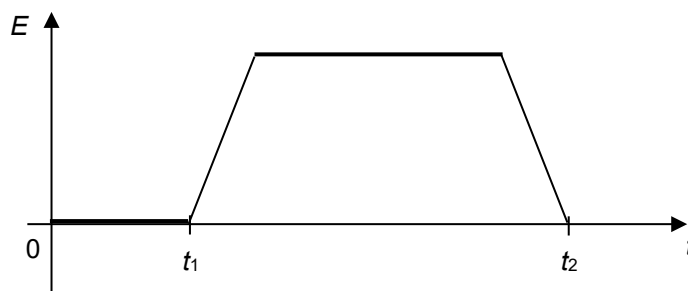
The e.m.f. induced in XY and the e.m.f. induced in WZ oppose each other. B1

Hence, the net e.m.f. is zero. B1

Marker's comments:

- When the loop is fully within the field, the induced e.m.f. is independent of v . The induced e.m.f. will still be zero if v is varying. Hence stating that v is constant does not explain why e.m.f. is zero.
- E.m.f. creates the force that drives the electrons in a circuit. Hence when induced e.m.f. in XY and WZ oppose each other, the force that drives the electrons cancel out. Hence it is less correct to state that the current due to e.m.f. in XY cancels out the current due to e.m.f. in WZ.

(ii)



1 mark for showing the linear increase and decrease of E . B1

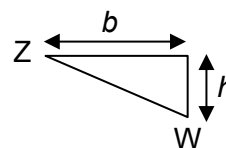
1 mark for the constant E . B1

Note:

For a triangle, area $A = \frac{1}{2}bh = \frac{1}{2}(kh)h = \frac{1}{2}kh^2$

$$E = \frac{d(NBA)}{dt} = NB \frac{dA}{dt} = NB \frac{d\left(\frac{1}{2}kh^2\right)}{dt} = \frac{1}{2}NBk \frac{dh^2}{dt} = \frac{1}{2}NBk \left(2h \frac{dh}{dt}\right) = NBkhv$$

Hence, E is proportional to h when b is not constant.



(iii) By Lenz's law, current is induced to produce a magnetic force to oppose the motion of the loop. Hence, an external force is needed so that there is no net force. B1

Marker's comments:

- Stating induced current is opposing the motion is incomplete as it is the magnetic force produced by the current in the magnetic field that opposes the motion and not current alone.
- It is important to state that there is induced current instead of just induced e.m.f. as induced e.m.f. does not necessarily produce a magnetic force if there is no complete circuit (e.g., moving rod in (a))

20 (a) (i)
$$\langle V \rangle = \frac{(5.0 \times 1.0) + (-7.5 \times 3.0) + (10.0 \times 2.0)}{6.0}$$
 M1

$$= 0.417 \text{ V}$$
 A1

Marker's comments: Students need to know clearly how to calculate the mean value. The mean value is the area under the graph over one cycle, divided by the time of one cycle (or period).

(ii)
$$V_{\text{rms}} = \sqrt{\frac{(5.0^2 \times 1.0) + ((-7.5)^2 \times 3.0) + (10.0^2 \times 2.0)}{6.0}}$$
 M1

$$= 8.10 \text{ V}$$
 A1

(iii)
$$\langle P_1 \rangle = \frac{V_{\text{rms}}^2}{R} = \frac{8.1^2}{28}$$
 M1

$$= 2.34 \text{ W}$$
 A1

Marker's comments: The mean power is given by the mean of the square of the voltage (and not the square of the mean), divided by the resistance of the resistor.

(b) For half-wave rectification of a sinusoidal voltage,

$$\begin{aligned} \langle P_2 \rangle &= \frac{1}{2} \times \frac{V_{\text{rms}}^2}{R} \\ &= \frac{1}{2} \times \frac{V_0^2}{2R} \\ &= \frac{1}{2} \times \frac{12^2}{2 \times 28} = 1.29 \text{ W} \end{aligned}$$
 M1
A1

Marker's comments: When a diode is present, there is a p.d. across the resistor only for half a cycle. That means that the power dissipated in the resistor is half that if there isn't a diode.

- 21 (a) A photon is a quantum/packet/discrete quantity B1
of electromagnetic energy/energy of electromagnetic radiation. B1

Marker's comments: A common mistake is not including "electromagnetic" but merely describing "energy". Some students made reference to the photoelectric effect, and it should be noted that most photons exist outside of the photoelectric effect. Hence, the photoelectric effect should not be used to define photons.
Some students defined proton instead of photon.

(b) (i) 1.
$$p = \frac{h}{\lambda}$$
 M1

$$= \frac{6.63 \times 10^{-34}}{650 \times 10^{-9}}$$
 A1

$$= 1.02 \times 10^{-27} \text{ kg m s}^{-1}$$

2.
$$E = \frac{hc}{\lambda}$$
 M1

$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{650 \times 10^{-9}}$$
 A1

$$= 3.06 \times 10^{-19} \text{ J}$$

(ii)
$$I = \frac{P}{S} = \frac{\left(\frac{n}{t}\right)E}{\pi r^2}$$
 C1

$$\left(\frac{n}{t}\right) = \frac{I\pi r^2}{E} = \frac{(4.5 \times 10^3)(\pi)(0.70 \times 10^3)^2}{3.06 \times 10^{-19}} = 2.26 \times 10^{16} \text{ s}^{-1}$$

B1

Marker's comments: Some students had difficulties with the manipulation of formula or getting the relationship between intensity and the rate of incidence of photons.

- (iii) Change in momentum of a photon during reflection,

$$\Delta p = p - (-p)$$

$$= 2p$$

$$= 2(1.02 \times 10^{-27})$$

$$= 2.04 \times 10^{-27} \text{ kg m s}^{-1}$$

C1

Force exerted on mirror = rate of change of momentum

$$= \left(\frac{n}{t}\right) \Delta p = (2.2637 \times 10^{16})(2.04 \times 10^{-27})$$

M1

$$= 4.62 \times 10^{-11} \text{ N}$$

A1

Marker's comments: Many students did not account for the change in momentum ($2p$) but instead used just the momentum of the photon, p . Since this is a reflection in the mirror, the photons must be travelling in the opposite direction after reflection, also at the speed of light.

- 22 (a) Nuclear fusion occurs when two light nuclei/nuclei of low mass number combine to form a nucleus of greater mass/larger mass number. B1

Marker's comments: A common mistake is to state "two *smaller* nuclei form a *larger* nucleus". "smaller" and "larger" are references to size and a reference to mass is required.

- (b) Energy released = (mass of deuterium and tritium) – (mass of helium and neutron)

$$= (3.01550 + 2.01355) - (4.00150 + 1.00866)$$

M1

$$= 0.01889 \text{ u}$$

$$= \frac{0.01889 \times 1.66 \times 10^{-27}}{1.60 \times 10^{-19} \times 10^6} \times (3.00 \times 10^8)^2$$

M1

$$= 17.6 \text{ MeV}$$

A1

- (c) This reaction has the highest rate of reaction at a lower temperature/temperature less than $10 \times 10^9 \text{ K}$. B1

Marker's comments: A common mistake is to mention the rate of reaction without any reference to the temperature. The deuterium-tritium reaction does not have the highest rate of reaction at $10 \times 10^9 \text{ K}$. Having the highest rate of reaction will not be useful if the temperature required is too high to be feasible.

- (d) Neutrons are electrically neutral and highly penetrative. B1

Marker's comments: A common mistake is to explain why it is necessary to stop neutrons from escaping the reactor.

The requirement for thick concrete implies that neutrons are highly penetrative. Answers that state that neutrons are highly penetrative is not given credit. An explanation of why neutrons are highly penetrative is required.

- (e) (i) The binding energy of a nucleus is the amount of energy required to separate to infinity all its constituent nucleons, B1

and is equal to the mass defect of the nucleus multiplied by c^2 , where c is the speed of light. B1

Marker's comments: A common mistake is to state that binding energy is the energy equivalence of mass defect. This answer does not state the relationship between binding energy and mass defect.

Another common mistake is to state $E = mc^2$, without defining what is E and m . When stating an equation, all the symbols used in the equation should be defined.

Another mistake is to state the binding energy is proportional to mass defect without stating what is the constant. The constant should be c^2 .

Some students provided the definition of mass defect or explained how to calculate mass defect when this is not required by the question.

$$\begin{aligned} \text{(ii) Total mass of nucleons} &= 15(1.00783)u + (31 - 15)(1.00866)u \\ &= 31.25601 u \end{aligned}$$

$$\begin{aligned} \text{Mass defect} &= 31.25601 u - 30.97376 u \\ &= 0.28225 u \end{aligned}$$

B1

$$\begin{aligned} \text{Binding energy per nucleon} &= \frac{0.28225 \times (3.0 \times 10^8)^2 \times 1.66 \times 10^{-27}}{31 \times 1.6 \times 10^{-19} \times 10^6} \\ &= 8.50 \text{ MeV} \end{aligned}$$

B1

A1

Marker's comments: Many students used the mass of a proton from the formula sheet as $1.67 \times 10^{-27} \text{ kg}$ instead of using the data given in the question. If this was followed by correct calculation, 2 marks was awarded.

Some students also mistakenly substituted the mass of an electron for the mass of a neutron. This would result in a maximum of 1 mark.

There were also students who forgot to calculate the binding energy per nucleon, and instead left their answers as binding energy. These answers were also awarded a maximum of 2 marks.

Some students rounded off the intermediate values too early, which resulted in the wrong final answer. Thus, the answer mark could not be awarded.

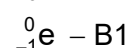
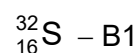
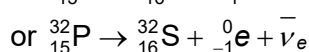
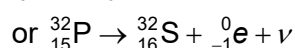
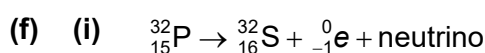
Students who did not calculate binding energy correctly (i.e., omitted the u or c^2 were not awarded the conversion mark from J to MeV)

Other common mistakes include calculating the energy released

- in the reaction of producing a phosphorus-31 atom from a phosphorus-32 atom or the other way around.
- just using the atomic mass of the phosphorus 31 atom instead of mass defect.

The presentation in this question could be improved. Students should state clearly what they are calculating. Many students

- only wrote down the formula, showed no substitution and immediately showed their final answer.
- omitted u from their working, only for it to magically be included in the final answer.
- did not bother to show the conversion from J to MeV, and instead just wrote one number and equated it into a different number in the line below without indicating a change in the unit.
- immediately converted it to MeV without showing the intermediate step of the answer in J.



neutrino – B1

Marker's comments:

- Many students did not recognise that a beta particle is an electron.
- Many students neglected to include the neutrino/anti-neutrino, even though they were aware that it is produced as evident from their answers in the next part of the question.

- Writing β instead of e is not accepted. Similarly, writing α instead of ${}^4_2\text{He}$ for an alpha particle is not accepted.
- Some students thought that a positron (${}^0_1\text{e}$) was produced instead of an electron.
- A good practice would be to check that the mass number and charge of the particles are conserved.

(ii) The energy released is shared between the kinetic energies of the sulfur B1 nuclide, beta particle and the anti-neutrino.

Marker's comments:

- Do note that there are three products for this reaction. Hence, the **energy released** is shared by **all** three and not just the beta particle and anti-neutrino. As such it is expected for students to mention all three products.
- Take note that it is not that the beta particle "*shares its kinetic energy*". The energy released is shared between the particles.
- Alternatively, if it is stated that the anti-neutrino carries away a range of kinetic energies, it is important to state "**range**" as it implies that there is variation in the distribution of energy released. Many students just stated that some energy is carried away. If this amount is fixed, there will not be a range of kinetic energy.
- Stating that the beta particle collides with other particle or describing its ionizing power and/or tracks through a cloud chamber is irrelevant as we are only interested in its kinetic energy due to its production from the decay of phosphorus to sulfur (before it goes on to ionize/collide with anything else).
- It is also insufficient to state that the variation of kinetic energy of the beta particle is due to the anti-neutrino possessing momentum due to the conservation of momentum. It is also insufficient to state that because the beta particle can move in different directions, it will have different kinetic energies. The links are not apparent, and more detail is needed to draw these links.
- Answers such as "because the beta particles are emitted with a range of kinetic energy" or "because they are emitted with different speeds" are paraphrasing the question and do not answer the question.
- Some students cited the random nature of radioactive decay for the range of kinetic energy. The random here does not refer to the products having random speeds. It refers to how each decay is random in that it is impossible to predict when the decay will occur.
- Many students also misinterpreted the question and tried to explain why the beta particle has a large kinetic energy without touching on the large **variation** of the energies produced.
- A significant number of students wrongly attributed this variation of kinetic energy to the Heisenberg's uncertainty principle or even the photoelectric effect, as well as electrons produced during X-ray production.

(iii) Decay constant λ = probability of decay per unit time. Hence, probability of decay in time Δt is given by $\lambda \Delta t$.

$$\therefore \lambda \Delta t = \frac{\ln 2}{14.3 \times 24} \times 2 = 0.00404$$

B1

Or

$$N = N_0 e^{-\lambda t}$$

$$\frac{\Delta N}{N_0} = \frac{N_0 - N}{N_0} = 1 - e^{-\lambda t} = 4.03 \times 10^{-3}$$

Marker's comments: Many students did not consider that the question asked for the probability in 2 hours and instead calculated decay constant in various denominations (per sec, per hour, per day).